

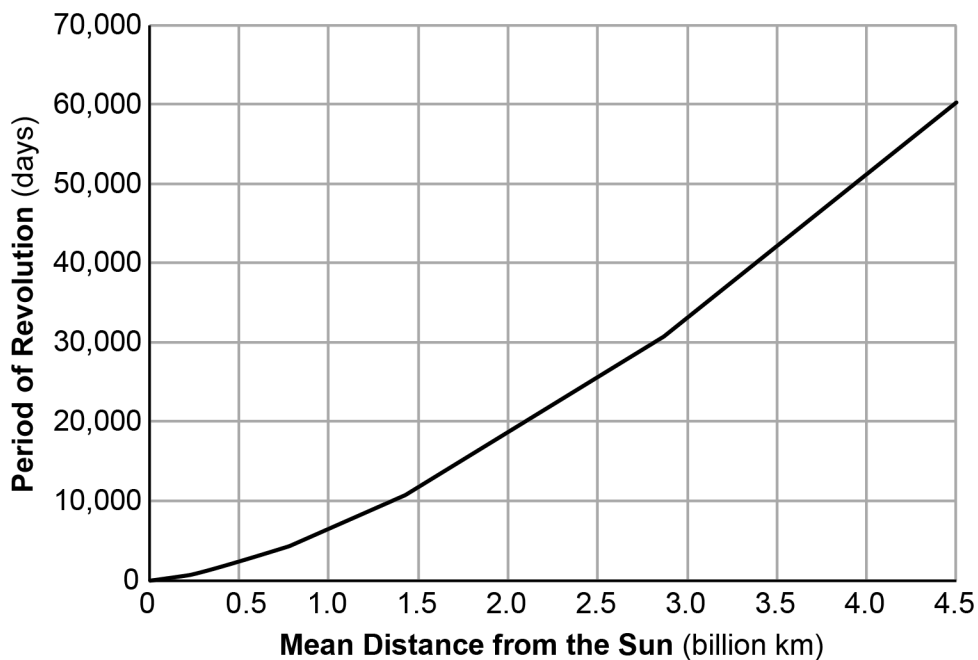
Name: _____

Unearthing Mars — A Historical Perspective

Student Answer Packet 2: To be completed individually.**QUESTIONS:**

1. Using your knowledge of Kepler's Laws and the graph below, make a claim using numerical evidence about the relationship between a planet's distance from the Sun and its period of revolution.

Relationship between Distance from the Sun and Period of Revolution



2. Describe how the location and motions of the planets in our solar system are governed by each of Kepler's Three Laws.

Law 1: _____

Law 2: _____

Law 3: _____

3. Based on the information from ***Student Answer Packet 1***, how were mathematical and computational representations used to predict the orbit and positions of Mars?

Orbit of Mars: _____

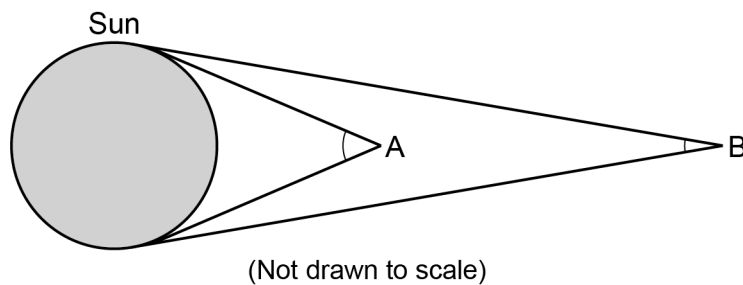
Positions of Mars: _____

4. Construct an explanation that supports the claim that perihelion and aphelion locations can be used to determine the length of a year (one orbit) for planets in our solar system.

5. How do gravitational effects between the Sun and a planet differ at the perihelion and the aphelion positions of the orbit of a planet?

6. The angular diameter of the Sun as viewed from Mars has been determined by the Curiosity Rover. Curiosity is a car-sized rover exploring geologic features on Mars. Angular diameter of the Sun is the apparent size of the Sun as viewed from the rover on Mars.

The model below represents two different distances and angular diameters of the Sun, as viewed from Mars, labeled A and B.



Explain why the apparent angular diameter of the Sun would be larger at perihelion. Use evidence from the model and numerical evidence from **Student Answer Packet 1** to justify your explanation.

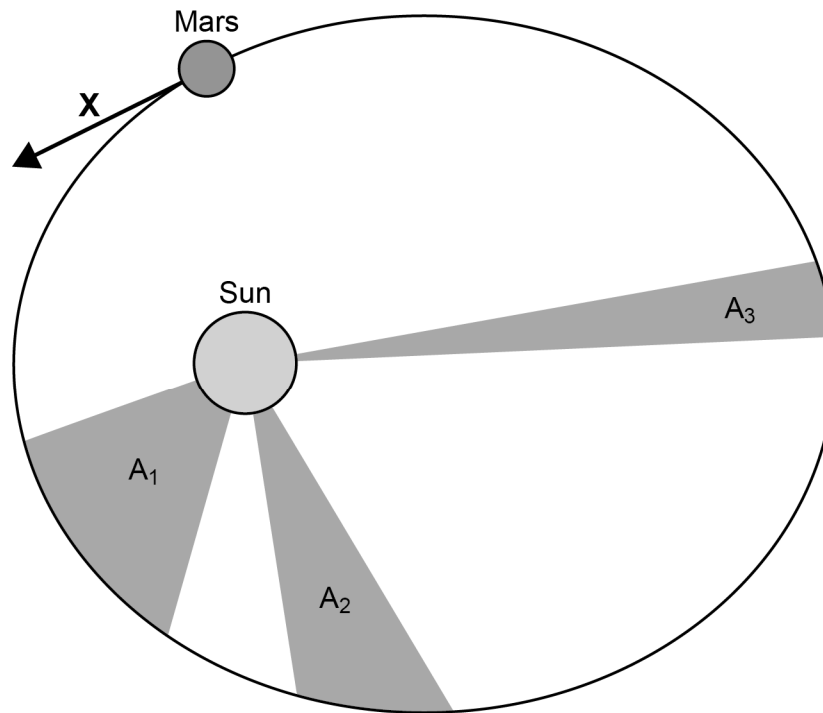
Model Evidence:

Numerical Evidence:

7. The arrow, labeled X, shown in the model below represents the path that Mars would take if the Sun were not present. Which force is responsible for the orbital path of Mars and all of the planets in our solar system?

8. Kepler's mathematical calculations and his theories were crucial to a better understanding of the scientific laws that govern our solar system. These discoveries, even though they were developed in the early 1500s, are still used to determine the locations of orbiting objects in our solar system and beyond.

The model below represents the path of Mars around the Sun. A_1 , A_2 , and A_3 are shaded regions of equal area. The distances traveled on the orbit along each of the three shaded areas are different, but the number of days it takes Mars to travel these distances is the same.



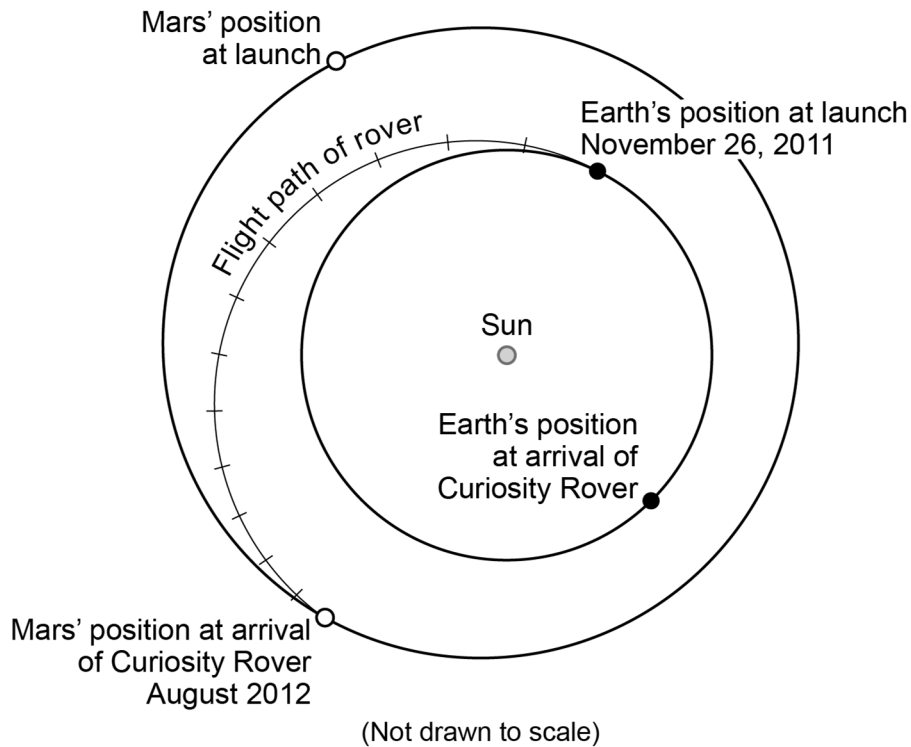
(Not drawn to scale)

Explain why Mars travels in its orbit for the same number of days as it passes along the edges of areas A_1 , A_2 , and A_3 even though the distances traveled are different.

9. Modern applications of Kepler's calculations are used today by NASA scientists. NASA has sent many satellites, landers, and rovers to Mars beginning with the first landing of Viking 1 on the surface of Mars in 1976. The model below represents the flight path for the Curiosity Rover after being launched from Earth on November 26, 2011.

The distance between tick marks on the flight path is 20 Earth days.

Flight Path of Curiosity Rover

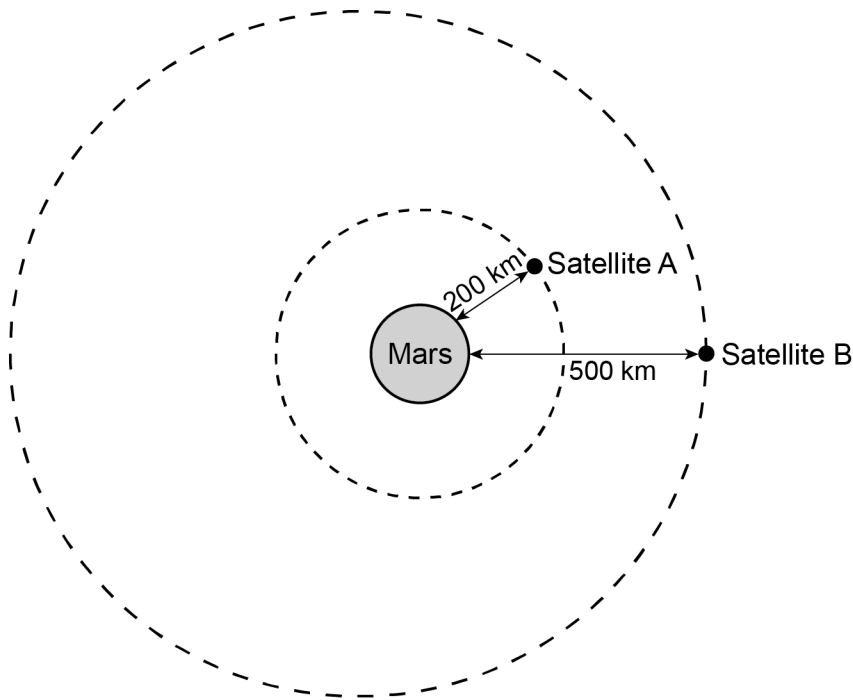


Use the model to calculate the approximate travel time between Earth and Mars in Earth days. Construct an explanation that describes why this flight path was selected rather than a straight-line path to Mars on the launch date.

a. Approximate travel time: _____ Earth days

b. Flight path explanation: _____

10. The model below represents two satellites, A and B, which have approximately the same mass. These satellites are orbiting Mars at different distances.



(Not drawn to scale)

Using your knowledge of the factors that affect gravitational attraction, describe the relative orbital speeds of *both* Satellites A and B. Give a reason for your answer using quantitative evidence from each satellite shown in the model.

Description of speeds:

Reason with quantitative evidence:

11. NASA has sent several different types of spacecrafts to Mars in the last 40 years. Each craft was designed for specific mission goals and to obtain different types of scientific information. The table below includes some information about some of the spacecrafts that have landed on Mars.

Selected Mars Spacecrafts

Spacecraft	Power Source	Cost (millions of dollars)	Approximate Length of Mission (years)	Mass (kilograms)
Spirit rover	solar panel	400	6	181.5
Phoenix Mars lander	solar panel	420	Less than 0.5	350.2
Curiosity rover	radioactive battery	2500	14	1025.1
Mars InSight lander	solar panel	829	4	360.2
Perseverance rover	radioactive battery	2400	14	1025.1

Engineers have implemented two different solutions to power sources for rovers and landers that are operating on Mars' surface. Each Mars rover and lander includes its own unique power system, based on the constraints engineers consider when designing the power systems for each particular Mars mission. These constraints include cost, safety, reliability, and environmental impact.

Evaluate and compare the **two** different power sources by describing **one** benefit and **one** drawback of *each* power source used on Mars spacecraft based on one criteria found in the table.

Solar panel benefit: _____

Solar panel drawback: _____

Radioactive battery benefit: _____

Radioactive battery drawback: _____

12. The Phoenix Mars lander landed on May 25, 2008 at 68°N latitude on Mars. The tilt of Mars' axis of rotation is approximately 25 degrees. Days on Mars (called Sols) are 24 hours, 37 minutes, 22 seconds.

The data table below shows information for Arctic Village, Alaska, which is located at 68°N latitude on Earth.

Arctic Village, Alaska			
Date	Sunrise	Sunset	Length of time Sun is above horizon
April 1	6:50 a.m.	8:43 p.m.	13 h 52 min
May 1	4:37 a.m.	10:44 p.m.	18 h 6 min
June 1	N/A	N/A	24 h
July 1	N/A	N/A	24 h
August 1	3:55 a.m.	11:37 p.m.	19 h 41 min
September 1	6:07 a.m.	9:13 p.m.	15 h 6 min

Construct an explanation to support the claim that the pattern of seasonal variations *and* the length of daylight hours at the Phoenix lander site on Mars and at Arctic Village, Alaska, are similar. Include **both** factors in your explanation.
